L-C-001804(DSA-DM)
June 12, 1972

INTERIM FEDERAL SPECIFICATION

CUSHION, RING, INFLATABLE, PLASTIC

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers an oval, inflatable ring cushion for use by medical patients who experience painful pressure when in the sitting or reclining position.

2. APPLICABLE DOCUMENTS.

2.1 Specification and Standards. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

L-P-375	Plastic Film, Flexible, Vinyl Chloride.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-76	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

FSC 6530

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Denver, San Francisco, Auburn, Washington, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and the Index of Federal Specifications from established distribution points in their agencies.)

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Vinyl sheeting shall comprise the cushion body and shall be 0.017 inch thick with an embossed surface. Vinyl shall not be compounded with any raw materials known to be skin sensitizers or irritants. The sheeting shall have the following physical properties:

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Thickness 0.017 inch ± .001 inch when measured in accordance with 4.4.1

Tensile Strength, nominal, psi	
Machine direction	2700
Transverse direction	2400
Minimum	2200

Ultimate Elongation, nominal, %	
Machine direction	300
Transverse direction	320
Minimum	280

Tensile strength and ultimate elongation shall be measured in accordance with 4.4.1.

Bondability shall be in accordance with 4.4.1. Exterior surface of the vinyl sheeting shall be airtight with no pin holes. The sheeting shall be tested for air tightness as described in 4.4.2. Any evidence of air bubble formation shall be cause for rejection. An acceptable sheeting is available as indicated in 6.4.

3.2 Construction. The cushion shall consist of vinyl sheeting and a retractable valve. Construction shall be in accordance with Figure 1. When tested as specified in 4.4.3 and 4.4.4, the cushion shall not burst, leak or otherwise show weakness.

3.2.1 Cushion body. The cushion body shall be vinyl sheeting in conformance with 3.1. Cushion seams shown in Figure 1 shall be uniform and smooth. The cushion shall be capable of being cleaned by washing with soap and water. Color shall be blue-green (see 6.4).

3.2.2 Valve. A retractable valve providing for oral inflation or pump connection shall be heat sealed to the cushion body and located as shown in Figure 1. The valve shall have an integral plug closure and shall easily close and roll flush into the inflated cushion to safeguard against accidental deflation and injury. The strength of the heat-sealed bond between valve and cushion body shall be at least equal to that specified in L-P-375 for bondability of two layers of polyvinylchloride (PVC) film. Paragraph 4.3.3 of L-P-375, as applicable, shall be used in testing bondability between valve and cushion body, and Table II of this specification shall apply. An acceptable valve is available as indicated in 6.5.

3.3 Identification marking. The manufacturer's name or registered trade mark shall be molded into the cushion body.

3.4 Workmanship. Workmanship shall be first class throughout. Ring cushions shall be free from defects which detract from their appearance or which may impair their seviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, suppled by the manufacturer of the vinyl sheeting may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the Federal Stock number, the Item Identification, the name of the component or material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Classification of Inspections.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and Table I.

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Table I.

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	S-2	1.0
Minor defects	S-2	4.0
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted on each lot of material in accordance with MIL-STD-105 and Table II.

Table II.

Component (unit of product)	Characteristic	Req.	Test Procedure	Sample Unit	Insp. Level
Vinyl sheeting	Physical properties	3.1	4.4.1	3 sq. yd.	Per lot of sheeting S-1
	Air tightness	3.1	4.4.2	1 complete cushion	
Complete cushion	Bursting strength	3.2	4.4.3	1 complete cushion	2 cushions per contract
	Leakage	3.2	4.4.4	1 complete cushion	2 cushions per contract
Valve	Bondability	3.2.2	See 3.2.2	1 complete cushion	2 cushions per contract

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4.3 Examination. The ring cushion shall be examined to determine compliance with this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects in Table III.

Table III.

Categories	Defects*
Major.	
101	Vinyl sheeting not free of cuts, cracks, nor tears.
102	Seams not completely heat sealed.
103	Vinyl sheeting not free of pin holes, patches nor imbedded foreign matter.
Minor	
201	Heat seals not free of wrinkles nor bunching.
202	Marking missing nor legible.
203	Vinyl sheeting not free of dirt, grease nor of loose foreign matter.
204	Cushion not the specified color.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The ring cushions shall be examined for defects in dimensions. Any dimension not within the tolerances specified in Figure 1 shall be classified as a defect.

4.4 Tests. Tests shall be conducted on each lot of material used in the fabrication of the cushion, as well as the finished cushion, to determine compliance with specification requirements.

4.4.1 Vinyl sheeting. Tests on the physical properties of the vinyl sheeting shall be conducted in accordance with the applicable procedures outlined in L-P-375 as follows:

Physical Property	Paragraph in L-P-375
Thickness	4.4.2
Tensile Strength	4.4.3
Ultimate Elongation	4.4.3
Bondability	3.4

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4.4.2 Air Tightness. A complete inflated ring cushion with the valve closed shall be completely submerged in water for a minimum of one minute. A record shall be kept on the location of any leaks through the cushion body as indicated by the formation of air bubbles.

4.4.3 Bursting strength. An evenly distrubuted load of 300 pounds shall be suddenly applied to the fully inflated cushion. From initial contact to full load shall not exceed half a second.

4.4.4 Leakage. Test requirement in 4.4.3 shall apply except the load shall be 200 pounds applied for at least 8 continuous hours.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing Twelve cushions, as specified constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	6 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each cushion shall be neatly folded and inserted into a plastic bag of appropriate size. The bag shall be closed with pressure sensitive adhesive tape. Twelve individually packaged cushions shall be further packaged in a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, type CF, class domestic. Commercial colors are acceptable on unit packages and the closure shall be adequate to prevent spilling of contents under normal handling.

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5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Levels A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. In addition, each exterior container shall be waterproofed in accordance with 30.4 of PPP-B-636. Special requirement of applicable table shall apply for PPP-B-636 box.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure shall conform to method II as specified in the appendix of PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The ring cushion is intended for use as an invalid ring designed to provide circumferential support for patients with disorders in rectal and adjacent areas.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).

6.3 This specification covers the following item appearing in the Federal Supply Catalog:

Federal Stock No.	Item Identification
6530-444-3375	CUSHION, RING, INFLATABLE, Plastic

6.4 Vinyl sheeting referenced in 3.1 is available from Union Carbide Corporation as their KDA2558-GN1675, with GN1675 indicating their shade of blue-green.

6.5 The valve referenced in 3.1.2 is available from Halkey-Roberts Corporation as their 450-AC Roll-In Valve.

6.6 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6530-1254

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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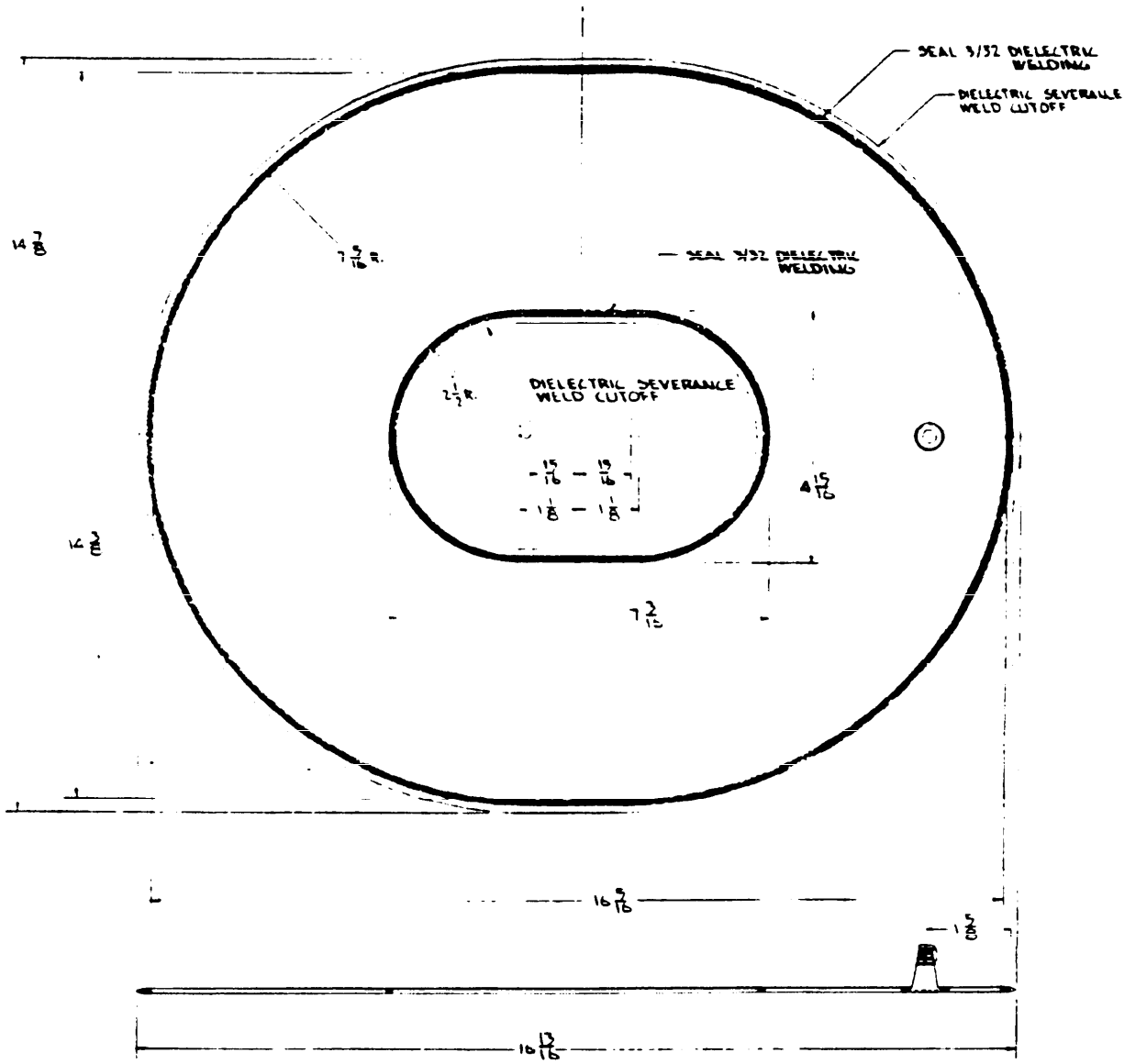


FIGURE 1

NOTE:
INFLATED SIZE: 7 1/2 x 17 x 17 1/2 OUTSIDE
5 1/2 x 8 1/4 INSIDE

TOLERANCES:
6 INCHES OR LESS ± 1/32 INCH
OVER 6 INCHES ± 1/64 INCH

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L-C-001804 (DSA-DM)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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